Note: Continuous preserves the Compactness, so, a Continuous map $f: X \to \mathbb{R}$ on Compact X has minimum and maximum, Since $f[X]$ is closed bounded.

「Darboux's Theorem.

Let $f: [a,b] \to \mathbb{R}$ be a differentiable map.

Then, the derivative f' satisfies the Intermidate Value Theorem.

Proof. Suppose that $f'(a) < f'(b)$, Let α s.t $f'(a) < \alpha < f'(b)$ be given.

Define $\quad g(x) = f(x) - \alpha x \quad (x \in [a,b])$.

Clearly g is differentiable, and $g'(a) = f'(a) - \alpha < 0$, $g'(b) = f'(b) - \alpha > 0$.

And, since g is continuous, it has a minimum in $[a,b]$, for some c.

By the assumption, $c \neq a$ and $c \neq b$. And, since g has a minimum at c, $g'(c) = f'(c) - \alpha = 0$. So proved ☒

「Corollary. If $f: I \to \mathbb{R}$ has a Primitive, then $f[I]$ is an interval.

Note: If f is defined on an interval and cannot satisfy IVT, then it has no Primitive.